

Subsets with Fixed Average

Time limit: 1.00 s

Memory limit: 512 MB

You are given an array of n integers. Your task is to count the number of non-empty subsets of the given array with average equal to a .

Input

The first line has two integers n and a : the size of the array and the target average.

The next line has n integers $x_1, x_2, \dots, x_n$: the contents of the array.

Output

Print one integer: the number of non-empty subsets with average equal to a , modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 500$
- $1 \leq a \leq 500$
- $1 \leq x_i \leq 500$

Example

Input	Output
5 2 1 1 2 3 4	7